CS 6515/4540: Graduate Algorithms

Fall 2026 – Exam 2 – Section B

October 27, 2026 Time: 70 minutes Total: 100 points

Instructors: Sahil Singla and Gerandy Brito

Name: _____

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INSTRUCTIONS

1. Write your name, GTID, or GT username on every page.
2. Bringing two pages of notes is permitted. No books, calculators, or other electronic devices are permitted.
3. This examination contains **8 (eight) pages**, including this cover. Verify that no pages are missing.
4. Write only on the **FRONT** of each sheet. Backs will not be scanned.
5. Make answers clear and concise. For Question 1, give a short justification unless the part says otherwise.
6. You may refer to standard algorithms and theorems from class without giving their full pseudocode or proofs, unless a part asks for a proof.
7. Parts are graded separately. You may use a stated claim from an earlier part even if you did not prove it. Show your reasoning for partial credit.
8. Unless stated otherwise, arithmetic and comparisons take $\mathcal{O}(1)$ time, arrays use indices starting at 1, and log means $\log_2$.

Question	Points
1. MCQs and Short answers	40
2. Approximation Algorithm	15
3. Convex Optimization and Gradient Descent	20
4. Deterministic Rounding	25
Total	100

I am aware of and in accordance with the Academic Honor Code of Georgia Tech and the Georgia Tech Code of Conduct. I will use no person's help on this test. I have read all the instructions on this page.

Signature: _____

1 Short Answer Questions (40 points)

Point values are shown for each part. Give a short justification (1–2 lines unless a proof or calculation is requested); a counterexample suffices for a false statement.

(a) (5 points) Consider a feasible primal maximization LP and its feasible dual minimization LP. Select ALL statements that are always true.

- ◯ The primal objective value is at most the dual objective value.
- ◯ If a primal-feasible solution and a dual-feasible solution have equal objective values, both are optimal.
- ◯ If x^* is an optimal solution of the primal, every constraint (of the primal) is tight.
- ◯ If the primal has a finite optimal value, the dual has an optimal solution with the same value.

1, 2, and 4.

1 is weak duality, 2 follows from weak duality, 4 is strong duality. 3 is false, imagine you have a constraint $x_1 \geq 0$ and also a constraint $x_1 \geq -1$, it could have an optimal solution but the latter constraint can never be tight.

(b) (5 points) Suppose a primal constraint holds strictly at an optimal primal solution x^*. What can we conclude about the corresponding variable in any optimal dual solution?

- ◯ It must equal zero.
- ◯ It must be strictly positive.
- ◯ It can be either positive or zero.
- ◯ The primal optimum cannot be unique.

1, this is complementary slackness.

(c) (6 points) Consider the linear program

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b, \end{aligned}$$

where $x \in \mathbb{R}^n$ is unrestricted in sign.

Let x^* be a feasible solution, and let S denote the set of constraints tight at x^*. Write a_i^T for row i of A.

Suppose there do **not** exist coefficients $\lambda_i \geq 0$ such that

$$c = \sum_{i \in S} \lambda_i a_i.$$

Select all statements that must be true.

- ◯ There exists a direction v such that $c^T v > 0$ and $a_i^T v \leq 0$ for all $i \in S$.
- ◯ There exists a direction v and $\varepsilon > 0$ such that $x^* + \varepsilon v$ is feasible and has strictly larger objective value.
- ◯ The linear program must be unbounded.

◯ Every constraint not in S at x^* remains satisfied under all sufficiently small perturbations of x^*.

1, 2, and 4.

1 is Farkas' Lemma

2 follows from 1 and 4. 1 gives us a v and says that for any $\varepsilon > 0$, $x^* + \varepsilon v$ still satisfies all the constraints in S and has strictly larger objective value. 4 gives us that if we choose ε to be small enough, then $x^* + \varepsilon v$ will also satisfy all constraints not in S. Therefore, the point will be feasible.

3 is not true. This will always hold at any non-optimal feasible solution, and the existence of a non-optimal feasible solution does not mean the LP is unbounded.

4 is true because if we consider only the constraints not in S and replace the $\leq$'s with $<$'s, x^* is in the resulting set (by definition of not being tight). Since the resulting set is the intersection of finitely many open sets (open half-spaces), it is open, which is equivalent to saying the constraints are still satisfied for small enough perturbations of x^*.

Question 1 continued

(d) (10 points) Consider the constrained optimization problem

$$\begin{aligned} \min \quad & f(x_1, x_2) = \frac{1}{2}(x_1^2 + 4x_2^2) - 2x_1 - 4x_2 \\ & x_1 + x_2 \leq 2, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Starting at $x^{(0)} = (0, 0)$, take one gradient descent step with $\eta = 1$. If the resulting point is infeasible, find the feasible point closest to it in Euclidean distance. Show your work.

First, $\nabla f(0, 0) = (-2, -4)$. So, applying gradient descent with $\eta = 1$, we get $(0, 0) - 1 \cdot (-2, -4) = (2, 4)$. This is outside the feasible region, because $2 + 4 = 6 > 2$.

The closest point to $(2, 4)$ that satisfies $x_1 + x_2 \leq 2$ is the point $(0, 2)$, since the closest point on a line to another point is just the perpendicular projection of that point that line. This also satisfies the other constraints, so $x^{(1)} = (0, 2)$.

Question 1 continued

(e) (6 points) Let $f : K \to \mathbb{R}$ be a differentiable and convex function defined on a convex domain K. Let $x \in K$ such that for any $y \in K$,

$$\langle \nabla f(x), y - x \rangle \geq 0$$

Prove that x is a global minimizer of f.

Fix any $y \in K$. By convexity and differentiability, we know

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \geq f(x) + 0 = f(x)$$

So, $f(y) \geq f(x)$ for any $y \in K$, meaning x is a global minimizer of f.

(f) (8 points) Suppose a greedy 2−approximation algorithm for Vertex Cover returns a vertex cover of size 20, and you know that the graph contains a matching of size 8.

What are the strongest lower and upper bound on the size of the minimum vertex cover that follows from this information? Briefly justify your answer.

We have an upper bound of 20 (there is a vertex cover of size 20, so the minimum can't be worse), and a lower bound of 10, since it is a 2-approximation.

The matching gives a lower bound of size 8 (any vertex cover must contain at least one endpoint of each edge in the matching), but this is weaker than the constraint from the vertex cover. Since the matching is not necessarily maximal, we cannot give an upper bound of 16 by taking all endpoints of the edges in the matching. So, the matching gives us no information, and our best bounds are [10,20].

2 Approximation Algorithm - Greedy (15 points)

Let $G = (V, E)$ be an undirected graph with maximum degree $\Delta \geq 1$. An *Independent Set (IS)* is a set $S \subseteq V$ of vertices such that no edge $e \in E$ connects two vertices in S. The goal is to find an IS of maximum size.

Consider the following greedy algorithm for Maximum Independent Set:

Algorithm 1 Greedy Independent Set

```
S ← ∅
while V(G) ≠ ∅ do
    Choose an arbitrary vertex v ∈ V(G)
    S ← S ∪ {v}
    Remove v and all its neighbors from G
end while
return S
```

(a) (5 points) Prove that the algorithm returns an independent set.

Proof is by contradiction. Assume after this algorithm, S contains some u, v s.t. $\{u, v\} \in E$. Since only one vertex is added at a time, we must have added u first or v first. Without loss of generality, assume we added u first.

Then, when we added u, we know v was not in S (by definition of first). Since v is a neighbor of u, it is then removed from G, meaning it cannot be added to S in the future. So, at the end of the algorithm $v \notin S$.

So, we have a contradiction, so the algorithm returns an independent set. □

(b) (8 points) Prove that the algorithm is a Δ-approximation for Maximum Independent Set.

Hint: Consider how many vertices are removed in each iteration and how it bounds the size of the set the algorithm returns.

Note, we can prove a $(\Delta + 1)$-approximation by just by noting that every time we add a vertex to S, we remove at most $\Delta + 1$ vertices from G, so the size of our approximation is at least $\frac{|V|}{\Delta+1}$, and any independent set is a subset of V.

But, I will prove the Δ-approximation as well. Define N_v as the closed neighborhood of v (contains v and all its neighbors). Note the size of N_v is at most $\Delta + 1$, as it contains only v and at most Δ neighbors.

Note that any independent set can only have at most Δ elements of N_v. This is because if the set contains v, it cannot contain any other elements of N_v, so it cannot contain $\Delta + 1$ elements. Also, since G is empty at the exit of our algorithm, all vertices in G are in at least one N_v.

So, any independent set can have size at most Δ times the number of such neighborhoods. Since the number of such neighborhoods is exactly the size of S, any independent set (and in particular the maximum independent set) can have size at most $\Delta|S|$, meaning the algorithm gives a Δ-approximation. □

(c) (2 points) Construct a family of graphs for which the algorithm can return an independent set of size exactly OPT/Δ.

Describe the graph family and explain which choices the greedy algorithm must make.

For any Δ, consider the "star graph" with Δ+1 vertices. This is a graph with one distinguished vertex which we will call v. Every other vertex is connected to v and nothing else. This graph is also know as $K_{1,\Delta}$, or the complete bipartite graph with sets of size 1 and Δ, respectively.

Note v has degree Δ and every other vertex has degree 1, so the maximum degree is Δ.

Now, the true maximum independent set is all vertices except v, which has size Δ. If this algorithm happens to choose v first, then since all other vertices are connected to v, the algorithm will terminate, giving an independent set of size 1, which is exactly OPT/Δ.

3 Convex Optimization (20 points)

Consider running gradient descent (unconstrained, over $\mathbb{R}^n$) on a convex, β-smooth function $f : \mathbb{R}^n \to \mathbb{R}$ where $\beta > 0$. We begin at $x_0 \in \mathbb{R}^n$ and we define our update to be

$$x_{k+1} = x_k - \frac{1}{\beta}\nabla f(x_k)$$

We will show a fact that was somewhat explored in the homework: that this gradient descent will make the gradient smaller as we go on.

(a) (8 points) Show

$$||\nabla f(x_{k+1})||_2 \leq ||\nabla f(x_k)||_2$$

i.e. the gradient decreases in 2-norm.

Hint: You may use the following fact without proof: For convex, β-smooth function f and any $x, y \in \mathbb{R}^n$, we have

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{\beta}||\nabla f(x) - \nabla f(y)||_2^2$$

Hint 2: In particular, try plugging in $x = x_k$ and $y = x_{k+1} = x_k - \frac{1}{\beta}\nabla f(x_k)$

We start by applying the hints, and I will make this longer than necessary by doing one step at a time (hopefully making it as clear as possible?).

$$\begin{aligned}
&\langle \nabla f(x_k) - \nabla f(x_{k+1}), x_k - x_{k+1} \rangle \geq \frac{1}{\beta}\left\|\nabla f(x_k) - \nabla f(x_{k+1})\right\|_2^2 \\
\iff &\left\langle \nabla f(x_k) - \nabla f(x_{k+1}), \frac{1}{\beta}\nabla f(x_k) \right\rangle \geq \frac{1}{\beta}\left\|\nabla f(x_k) - \nabla f(x_{k+1})\right\|_2^2 \\
\iff &\frac{1}{\beta}\langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_k) \rangle \geq \frac{1}{\beta}\left\|\nabla f(x_k) - \nabla f(x_{k+1})\right\|_2^2 \\
\iff &\langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_k) \rangle \geq \left\|\nabla f(x_k) - \nabla f(x_{k+1})\right\|_2^2 \\
\iff &\langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_k) \rangle \geq \langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_k) - \nabla f(x_{k+1}) \rangle \\
\iff &\langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_k) \rangle - \langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_k) - \nabla f(x_{k+1}) \rangle \geq 0 \\
\iff &\langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_k) - (\nabla f(x_k) - \nabla f(x_{k+1})) \rangle \geq 0 \\
\iff &\langle \nabla f(x_k) - \nabla f(x_{k+1}), \nabla f(x_{k+1}) \rangle \geq 0 \\
\iff &\langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle - \langle \nabla f(x_{k+1}), \nabla f(x_{k+1}) \rangle \geq 0 \\
\iff &\langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle - \left\|\nabla f(x_{k+1})\right\|_2^2 \geq 0 \\
\iff &\langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle \geq \left\|\nabla f(x_{k+1})\right\|_2^2
\end{aligned}$$

Now, we can use the Cauchy-Schwarz inequality to see

$$\left\|\nabla f(x_k)\right\|_2 \left\|\nabla f(x_{k+1})\right\|_2 \geq \langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle$$

So, by transitivity,

$$\left\|\nabla f(x_k)\right\|_2 \left\|\nabla f(x_{k+1})\right\|_2 \geq \left\|\nabla f(x_{k+1})\right\|_2^2$$

and after canceling, we get

$$\left\|\nabla f(x_k)\right\|_2 \geq \left\|\nabla f(x_{k+1})\right\|_2$$

□

(b) (6 points) Let $x^* \in \arg\min f(x)$. Prove that

$$||\nabla f(x_k)||_2 \leq \sqrt{\frac{2\beta}{k+1}\left(f(x_0) - f(x^*)\right)}$$

Hint: You may use $\sum_{i=0}^{k} ||\nabla f(x_i)||_2^2 \leq 2\beta\left(f(x_0) - f(x^*)\right)$ without proof. (This was a homework problem!)

First,

$$\sum_{i=0}^{k} \|\nabla f\left(x_i\right)\|_2^2 \geq (k+1) \min_{i=0..k} \|\nabla f\left(x_i\right)\|_2^2$$

(this is true for any summation of values, we can only decrease each element in the summation).

Since we showed in the previous part that $\|\nabla f(x_i)\|_2$ is non-increasing (and therefore its square is non-increasing, since norms are nonnegative), the minimum is attained at x_k, which gives

$$\sum_{i=0}^{k} \|\nabla f\left(x_i\right)\|_2^2 \geq (k+1) \|\nabla f\left(x_k\right)\|_2^2$$

By transitivity (using the equation from the hint),

$$(k+1) \|\nabla f\left(x_k\right)\|_2^2 \leq 2\beta\left(f(x_0) - f(x^*)\right)$$

If we divide both sides by $k+1$ and take a square root of both sides, we get the desired inequality. □

(c) (6 points) Fix some optimum $x^* \in \arg\min f$, and let $D = ||x_0 - x^*||_2$. Show

$$||\nabla f(x_k)||_2 \leq \frac{\beta D}{\sqrt{k+1}}$$

Hint: Use part (b) + definition of β-smooth function.

First, note that $\nabla f(x^*) = 0$, since f is a convex differentiable function on an open set (if it were not zero, if we take a small enough step in the opposite direction of the gradient, the function value would decrease).
Now, we can apply the quadratic upper bound of β-smoothness to see

$$f(x_0) \leq f(x^*) + \langle \nabla f(x^*), x_0 - x^* \rangle + \frac{\beta}{2} \|x_0 - x^*\|_2^2$$

Since $\nabla f(x^*) = 0$, the inner product term is 0, so rearranging gives

$$f(x_0) - f(x^*) \leq \frac{\beta}{2} \|x_0 - x^*\|_2^2 = \frac{\beta D^2}{2}$$

Since everything is nonnegative, we can substitute in $\frac{\beta D^2}{2}$ for $f(x_0) - f(x^*)$ in the inequality from the previous part, and it will still hold. Simplifying, we get exactly

$$|\nabla f(x_k)||_2 \leq \frac{\beta D}{\sqrt{k+1}}$$

□

4 Approximation Algorithms (30 points)

Let $U = \{u_1, \ldots, u_n\}$ be a universe of elements, and let $S_1, \ldots, S_m \subseteq U$ be subsets whose union is U. Assume that every element belongs to **at most k sets**, where $k \geq 1$. The goal is to select the minimum number of sets whose union is U.

For each set S_i, introduce a variable x_i indicating whether S_i is selected.

1. (6 pts) Write an integer linear program (ILP) for this problem.

$$\min \sum_{i=1}^{m} x_i$$

s.t.

$$\sum_{i \text{ s.t. } u_j \in S_i} x_i \geq 1 \qquad \text{for all } j \in \{1, ..., n\}$$

$$x_i \in \{0, 1\} \qquad \text{for all } i \in \{1, ..., m\}$$

2. (6 pts) Let x^* be an optimal solution to the LP relaxation. Consider the rounding rule

$$\widehat{x}_i = \begin{cases} 1 & \text{if } x_i^* \geq 1/k, \\ 0 & \text{otherwise.} \end{cases}$$

Prove that the collection $\{S_i \mid \hat{x}_i = 1\}$ covers U.

Fix any u_j. Let $r_1, ..., r_l$ be all the indices such that $u_j \in S_{r_i}$.

Now, by the constraint associated with u_j, we know

$$\sum_{i=1}^{l} x_{r_i}^* \geq 1$$

So, we also know

$$l \cdot \max_{i \in \{1,\ldots,l\}} x_{r_i}^* \geq 1$$

so there is some i s.t. $x_{r_i}^* \geq \frac{1}{l}$. Now, since l is the number of sets that contain u_j, we know it's at most k, so we also know that $x_{r_i}^* \geq \frac{1}{k}$ (for the same i), so $\hat{x}_{r_i} = 1$, so S_{r_i} is in the collection. Since we know $u_j \in S_{r_i}$, u_j is covered.

Since this argument works for all $u_j \in U$, the collection covers all of U.

3. (8 pts) Prove that the rounding algorithm is a k-approximation.

Define the value of the optimal solution to the relaxed LP to be LP^*, and for the original problem to be ILP^*. Since the LP is a relaxed version and this is a minimzation problem, we know

$$LP^* \leq ILP^*$$

Now, I will show the value of our rounded solution is at most $k \cdot LP^*$, which also means it's at most $k \cdot ILP^*$, making it a k-approximation to the original problem.

First, note that for any i,

$$\hat{x}_i \leq k x_i^*$$

This we can see with case analysis. If $\hat{x}_i = 0$, this is trivial, and if $\hat{x}_i = 1$, then $x_i^* \geq \frac{1}{k}$, so $k x_i^* \geq 1 = \hat{x}_i$.

So,

$$k \cdot LP^* = k \sum_{i=1}^{m} x_i^* = \sum_{i=1}^{m} k x_i^* \geq \sum_{i=1}^{m} \hat{x}_i$$

So, this is a k-approximation in LP value and thus the ILP value. For the ILP, the value is exactly the number of sets used to cover U, so it is a k-approximation to the original problem.

4. (5 pts) Construct an instance with $k = 3$ and an optimal fractional LP solution x^* for which the rounding algorithm selects three sets, while an optimal integral solution selects only one set.

Specify the universe, the sets, and the fractional solution.

Let $U = S_1 = S_2 = S_3 = \{1, 2, 3\}$. Note that $x^* = (1/3, 1/3, 1/3)$ is an optimal solution to the fractional LP, since it satisfies all the constraints, and any solution must have $x_1 + x_2 + x_3 \geq 1$, so we cannot have a value smaller than 1.

When we round according to the above scheme, everything rounds to 1, so all sets are taken. However, clearly S_1 alone covers U, so the optimal integral solution only has one set.

END OF EXAM